

Experiment 3: Camera Calibration using Checkerboard Images

Aim:

To estimate intrinsic and extrinsic camera parameters using checkerboard images.

Theory:

Camera calibration is the process of estimating the internal parameters of a camera (intrinsic) and its position in space (extrinsic). This is achieved by analyzing multiple images of a known pattern, such as a checkerboard. The intrinsic parameters include focal length and optical center, while extrinsic parameters include rotation and translation vectors that describe the camera's position relative to the pattern. OpenCV provides `cv2.calibrateCamera()` to compute these parameters by detecting checkerboard corners in a set of calibration images.

Algorithm:

- Start
- Import required libraries (`cv2`, `numpy`, `glob`, `os`)
- Define checkerboard size and prepare 3D object points
- Read all checkerboard images from a folder
- For each image:
 - - Convert to grayscale
 - - Detect checkerboard corners using `cv2.findChessboardCorners()`
 - - If found, refine corners and save object and image points
- Calibrate camera using `cv2.calibrateCamera()` to get intrinsic and extrinsic parameters
- Compute and print the mean reprojection error
- End

Flowchart:

Start → Import Libraries → Load Images → Convert to Grayscale → Detect Checkerboard Corners → If Corners Found → Refine & Store Points → Repeat for All Images → Calibrate Camera → Print Parameters → End

Viva Questions:

- What is the difference between intrinsic and extrinsic parameters?
- What does the reprojection error represent?
- Why is a checkerboard pattern commonly used for calibration?